Systems of Linear Equations

Concept

A system solution satisfies both equations at once: it is their graph intersection.

Substitution

Replace an isolated variable in the other equation.

Elimination

Make coefficients opposite, then add the equations.

Number of solutions

Intersecting: one; parallel: none; same line: infinitely many.

Check

Substitute the ordered pair into both original equations.

Example 1

Solve $y = 2x + 1$ and $x + y = 10$.

Solution

1. Substitute: $x + (2x + 1) = 10$.
2. $3x = 9 \rightarrow x = 3$; then $y = 7$.
3. Solution $(3,7)$; it satisfies both equations.

Systems of Linear Equations

Example 2

Solve $2x + 3y = 13$ and $3x - 3y = 12$.

Solution

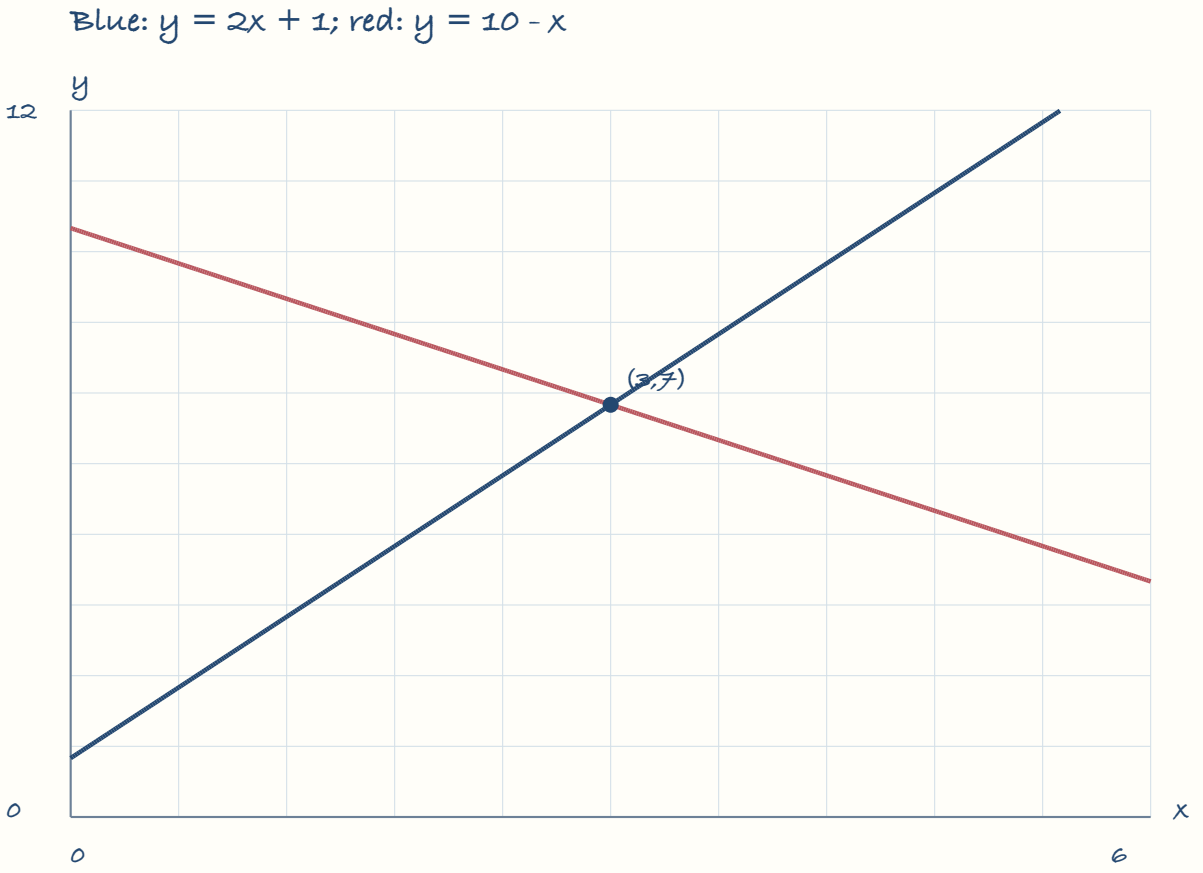
1. Add equations: $5x = 25 \rightarrow x = 5$.

2. Substitute: $10 + 3y = 13 \rightarrow y = 1$.

3. Solution $(5,1)$.

Systems of Linear Equations

Visual notes | read the graph



The intersection satisfies both equations. Substitution confirms the graph reading.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Systems of Linear Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $y = x - 2$ and $x + y = 8$.

2. Solve $3x + 2y = 16$ and $x - 2y = 0$.

3. Classify $x + y = 4$ and $2x + 2y = 8$.

4. Two plans cost $A = 10 + 2v$ and $B = 4v$. Find their break-even use and cost.

Systems of Linear Equations

Practice answers | check after trying

1. $2x - 2 = 8$ gives $x = 5$, $y = 3$.

2. Adding gives $4x = 16$, so $x = 4$ and $y = 2$.

3. Infinitely many solutions: the equations describe the same line.

4. $10 + 2v = 4v$ gives $v = 5$ visits; both cost $20. Below 5 visits B is cheaper; above 5 A is cheaper.